II YEAR I SEMESTER
DATA STRUCTURES

Regulation	Year-Sem	Course Code	Category	Periods/ Week			Credits	Maximum Marks		
				L	T	P	C	CIE	SEE	Total
R22	II – I	A63E1	PC	3	-	-	3	40	60	100

Prerequisites: Computational Thinking and Programming

Course Objectives:

1. Interpret the performance of linear and binary searching algorithms based on recursive functions and time, space complexities.
2. Implement the sorting, hashing and collision resolution techniques.
3. Develop linear data structures for stacks & queues using arrays.
4. Develop singly, doubly & circular linked list data structures.
5. Illustrate binary tree and binary search trees and Understand graph types, representations and graph traversal algorithms.

UNIT – I (Text book 1, Chapter 1, 2 & 13)
Algorithms: Preliminaries of algorithm, Algorithm analysis and complexity.
Recursion: Definition, Design Methodology and Implementation of recursive algorithms, types of recursion, recursive algorithms for factorial function, GCD computation, Fibonacci sequence, Towers of Hanoi, **Searching Techniques:** Linear Search, Binary Search.

UNIT – II (Text book 1, Chapter 12)
Sorting Techniques: Basic concepts, Insertion Sort, Selection Sort, Bubble Sort, Quick Sort, Merge Sort and Counting Sort. **Hashing:** Hash table representations, Collision resolution: separate chaining, open addressing, linear probing, quadratic probing, Double hashing, Rehashing.

UNIT – III (Text book 1, Chapter 3 & 4)
Stacks: Basic Stack Operations, Representation of a Stack using Arrays, Stack Applications: Reversing a list, Infix to postfix Transformation, Evaluating Postfix Expression. **Queues:** Basic Queues Operations, Implementation of Queue Operations using arrays, Application of Queues: Round Robin Algorithm, Circular Queues.

UNIT – IV (Text book 1, Chapter 5)
Linked Lists: Introduction, single linked list, representation of a linked list in memory, Operations on a single linked list, merging two single linked lists into one list, reversing a single linked list, advantages and disadvantages of single linked list, Circular linked list, Double linked list.

UNIT –V (Text book 1, Chapter 6, 7 & 11)
Trees: Basic tree concepts, **Binary Tree:** Properties, Representation of Binary Trees – Array Representation and Linked Representation, Binary Tree Traversals (recursive), Creation of binary tree from in-order and pre/post-order traversals, Binary Search Tree Operations and Implementation. **Graphs:** Definition, Properties, Representation of Un-Weighted Graphs, Weighted Graphs, Graph Search Methods: BFS, DFS, and Implementation.

Course Outcomes:

1. Interpret the performance of linear and binary searching algorithms based on recursive functions, time and space complexities. (BL-2)
2. Implement sorting, hashing and collision resolution techniques. (BL-2)
3. Develop linear data structures for stacks, queues using arrays. (BL-3)
4. Develop singly, doubly & circular linked list data structures. (BL-3)
5. Illustrate binary tree and binary search trees and Understand graph types, representations and graph traversal algorithms. (BL-3)

Text Books:

1. Data Structures, 2/e, Richard F, Gilberg , Forouzan, Cengage

References:

1. Data Structures using C, Aaron M. Tanenbaum, Yedidyah Langsam, Moshe J Augenstein, Pearson

2. Data structures and Algorithm Analysis in C, Mark Allen Weiss, Pearson Education. Ltd., Second Edition

II YEAR I SEMESTER
DATA STRUCTURES

Regulation	Year-Sem	Course Code	Category	Periods/ Week			Credits	Maximum Marks		
				L	T	P	C	CIE	SEE	Total
R22	II – I	A63E1	PC	3	-	-	3	40	60	100

Prerequisites: Computational Thinking and Programming

Course Objectives:

1. Interpret the performance of linear and binary searching algorithms based on recursive functions and time, space complexities.
2. Implement the sorting, hashing and collision resolution techniques.
3. Develop linear data structures for stacks & queues using arrays.
4. Develop singly, doubly & circular linked list data structures.
5. Illustrate binary tree and binary search trees and Understand graph types, representations and graph traversal algorithms.

UNIT – I (Text book 1, Chapter 1, 2 & 13)
Algorithms: Preliminaries of algorithm, Algorithm analysis and complexity. **Recursion**: Definition, Design Methodology and Implementation of recursive algorithms, types of recursion, recursive algorithms for factorial function, GCD computation, Fibonacci sequence, Towers of Hanoi, **Searching Techniques**: Linear Search, Binary Search.

UNIT – II (Text book 1, Chapter 12)
Sorting Techniques: Basic concepts, Insertion Sort, Selection Sort, Bubble Sort, Quick Sort, Merge Sort and Counting Sort. **Hashing**: Hash table representations, Collision resolution: separate chaining, open addressing, linear probing, quadratic probing, Double hashing, Rehashing.

UNIT – III (Text book 1, Chapter 3 & 4)
Stacks: Basic Stack Operations, Representation of a Stack using Arrays, Stack Applications: Reversing a list, Infix to postfix Transformation, Evaluating Postfix Expression. **Queues**: Basic Queues Operations, Implementation of Queue Operations using arrays, Application of Queues: Round Robin Algorithm, Circular Queues.

UNIT – IV (Text book 1, Chapter 5)
Linked Lists: Introduction, single linked list, representation of a linked list in memory, Operations on a single linked list, merging two single linked lists into one list, reversing a single linked list, advantages and disadvantages of single linked list, Circular linked list, Double linked list.

UNIT –V (Text book 1, Chapter 6, 7 & 11)
Trees: Basic tree concepts, Binary Tree: Properties, Representation of Binary Trees – Array Representation and Linked Representation, Binary Tree Traversals (recursive), Creation of binary tree from in-order and pre/post-order traversals, Binary Search Tree Operations and Implementation. **Graphs**: Definition, Properties, Representation of Un-Weighted Graphs, Weighted Graphs, Graph Search Methods: BFS, DFS, and Implementation.

Course Outcomes:

1.Interpret the performance of linear and binary searching algorithms based on recursive functions, time and space complexities. (BL-2)
2.Implement sorting, hashing and collision resolution techniques. (BL-2)
3.Develop linear data structures for stacks, queues using arrays. (BL-3)
4.Develop singly, doubly & circular linked list data structures. (BL-3)
5.Illustrate binary tree and binary search trees and Understand graph types, representations and graph traversal algorithms. (BL-3)

Text Books:

1. Data Structures, 2/e, Richard F, Gilberg , Forouzan, Cengage

References:

1.Data Structures using C, Aaron M. Tanenbaum, Yedidyah Langsam, Moshe J Augenstein, Pearson
2.Data structures and Algorithm Analysis in C, Mark Allen Weiss, Pearson Education. Ltd., Second Edition